Deep Learning

Language Models

Course Contents

Deep Learning

Transformer

LLMs

Context Engineering

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The Bigger Picture

This Part: Deep Learning

- NLP introduction
- neural networks (example: image classification)
- network training (with PyTorch)

Natural Language Processing (NLP)

field within AI and linguistics

aim: enable machines to understand and generate human language

language models: subfield of NLP

aim: model and understand language patterns (typically by learning to predict the next word in a sentence)

capabilities of large language models (LLM) cover most of NLP

NLP Tasks

- part-of-speech tagging
- named entity recognition
- text classification
- machine translation
- summarization
- question answering
- text generation (chatbots)
- code generation
- …

Symbolic vs Neural NLP

symbolic NLP = rules and logic

- uses hand-crafted rules, lexicons, and grammars
- relies on linguistic knowledge and logic-based systems

neural NLP = data and learning

- uses machine learning (especially deep learning)
- learns patterns from large datasets without explicit rules

Learning from Data

ImageNet

The Lord of the Rings

Article Talk

From Wikipedia, the free encyclopedia

(Redirected from Lord of the rings)

This article is about the book. For other uses, see The Lord of the Rings (disambiguation).
"War of the Ring" redirects here. For other uses, see War of the Ring (disambiguation).

The Lord of the Rings is an epic[1] high fantasy novel[a] by the English author and scholar J. R. R. Tolkien. Set in Middle-earth, the story began as a sequel to Tolkien's 1937 children's book *The Hobbit*, but eventually developed into a much larger work. Written in stages between 1937 and 1949, *The Lord of the Rings* is one of the best-selling books ever written, with over 150 million copies sold.[2]

The title refers to the story's main antagonist,[b] Sauron, the Dark Lord who in an earlier age created the One Ring to rule the other Rings of Power given to Men, Dwarves, and Elves, in his campaign to conquer all of Middle-earth. From homely beginnings in the Shire, a hobbit land reminiscent of the English countryside, the story ranges across Middle-earth, following the quest to destroy the One Ring, seen mainly through the eyes of the hobbits Frodo, Sam, Merry, and Pippin. Aiding Frodo are the Wizard Gandalf, the Men Aragorn and Boromir, the Elf Legolas, and the Dwarf Gimli, who unite in order to rally the Free Peoples of Middle-earth against Sauron's armies and give Frodo a chance to destroy the One Ring in the fire of Mount Doom.

Although often mistakenly called a trilogy, the work was intended by Tolkien to be one volume in a two-volume set along with *The Silmarillion*.[3][T 3] For economic reasons, *The Lord of the Rings* was first published over the course of a year from 29 July 1954 to 20 October 1955 in three volumes rather than one[3][4] under the titles *The Fellowship of the Ring*, *The Two Towers*, and *The Return of the King*; *The Silmarillion* appeared only after the author's death. The work is divided internally into six books, two per volume, with several appendices of background material.[c] These three volumes were later published as a boxed set, and even finally as a single volume, following the author's original intent.

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1	2	20	RL	80.0	9600	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	NaN	NaN	0	5	2007	WD	Normal	181500
2	3	60	RL	68.0	11250	Pave	NaN	IR1	Lvl	AllPub	...	0	NaN	NaN	NaN	0	9	2008	WD	Normal	223500
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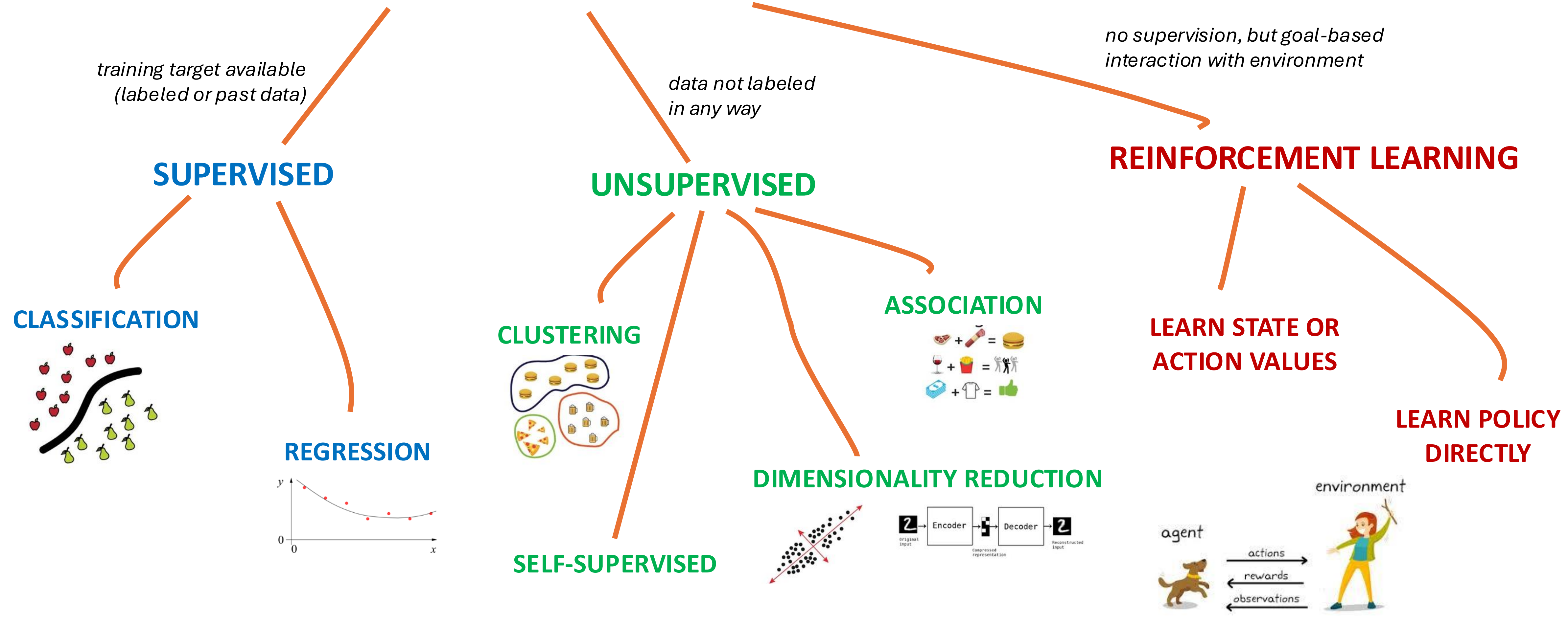
MACHINE LEARNING



learning by teacher (high-dimensional curve fitting)

learning by observation (pattern recognition)

learning by trial-and-error (sequential decision making)

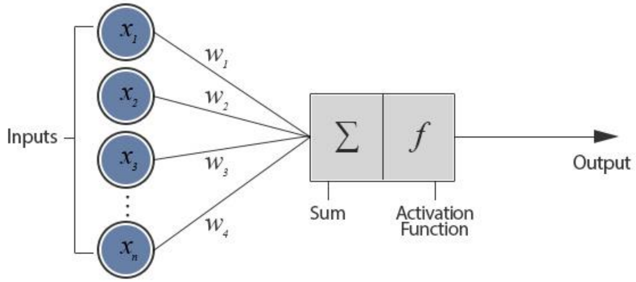
unsupervised and reinforcement learning can both be cast as supervised-learning setup

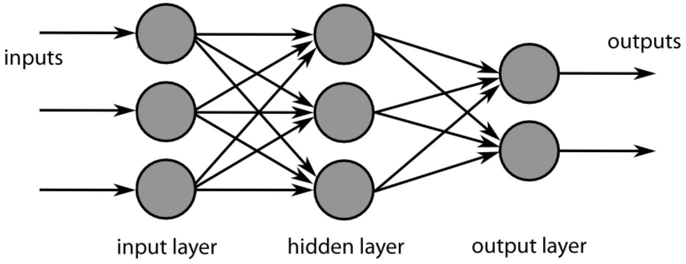
Algorithmic Families of Supervised Learning

parametric models

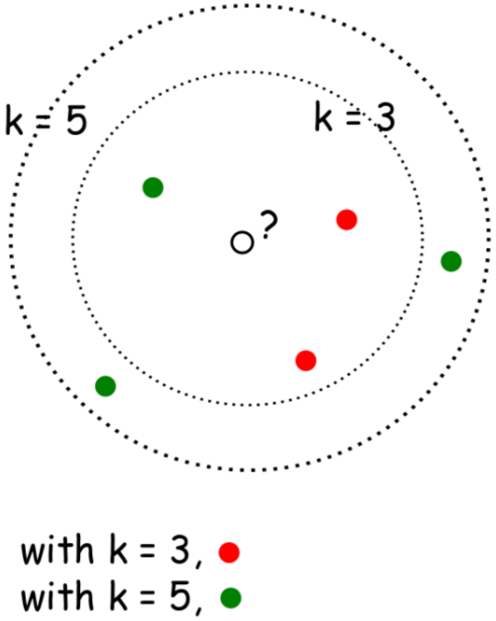
linear regression

neural networks: many linear models, nonlinear by means of activation functions



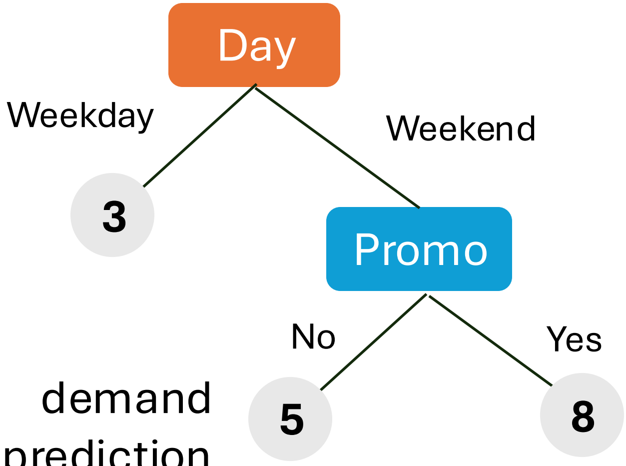


nearest neighbors (local methods, instance-based learning): non-parametric models



kernel/support-vector machines: linear model (maximum-margin hyperplane) with kernel trick

decision trees: rule learning



demand prediction

often used in ensemble methods

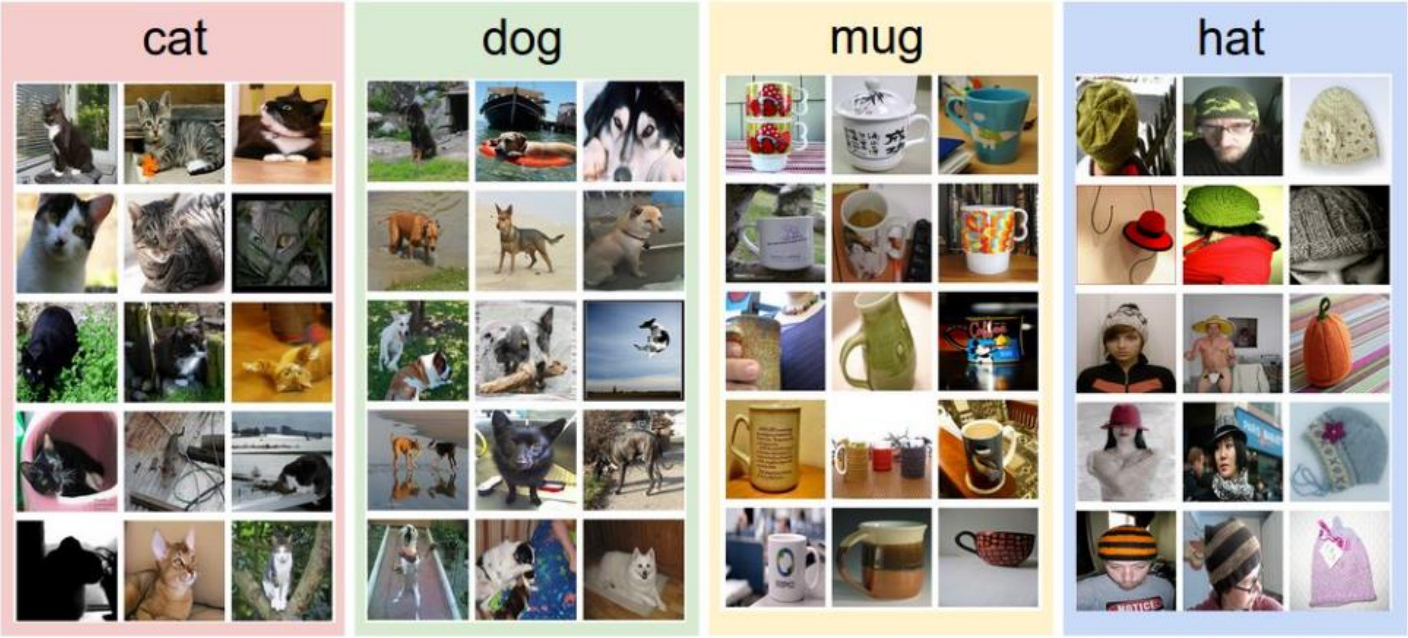
- bagging: random forests
- boosting: gradient boosting

Example: Image Classification

training data:

test data with learned model:

labels:



$f($  $) =$ "Cat"

$f($  $) =$ "Dog"

Image Classification as Input-Output Mapping

$f($  $) =$ “Cat”

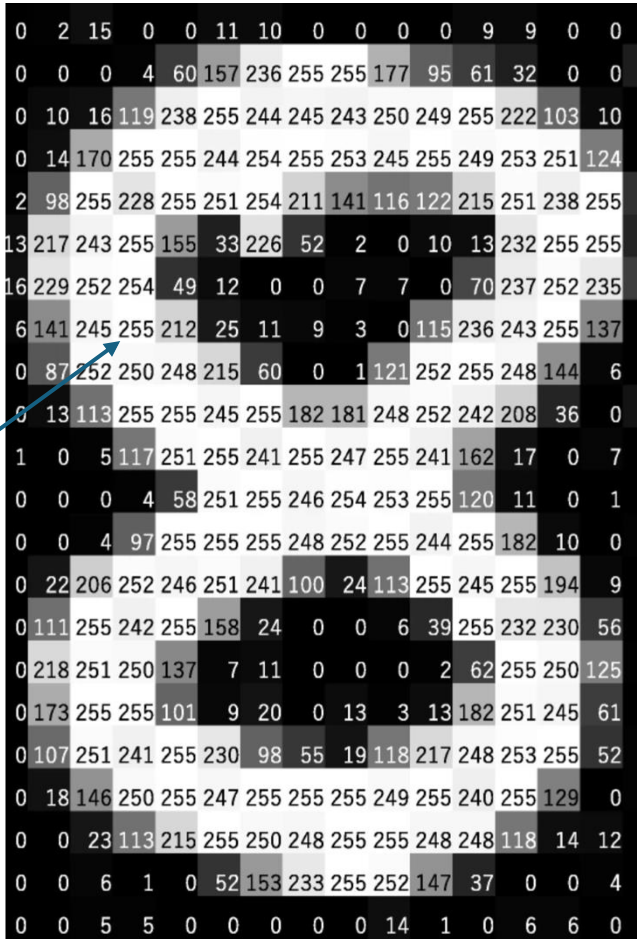
complicated function
→ neural network

$f($  $) =$ “Dog”

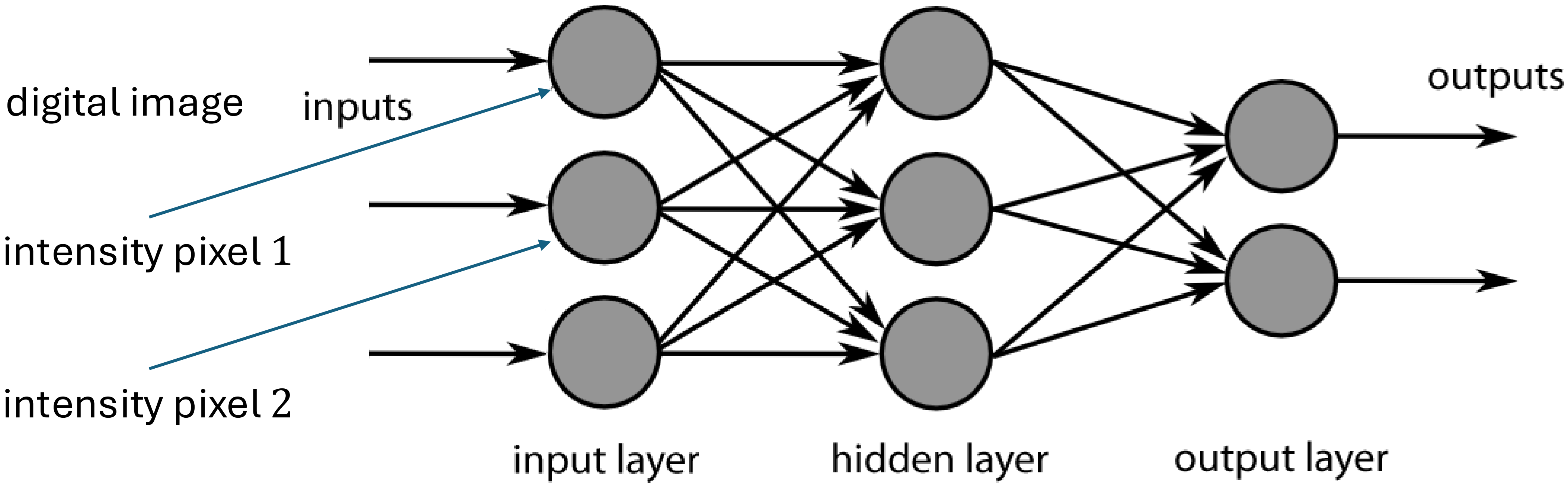
How Computers See Images

digital images:
pixels und intensity

8-bit image

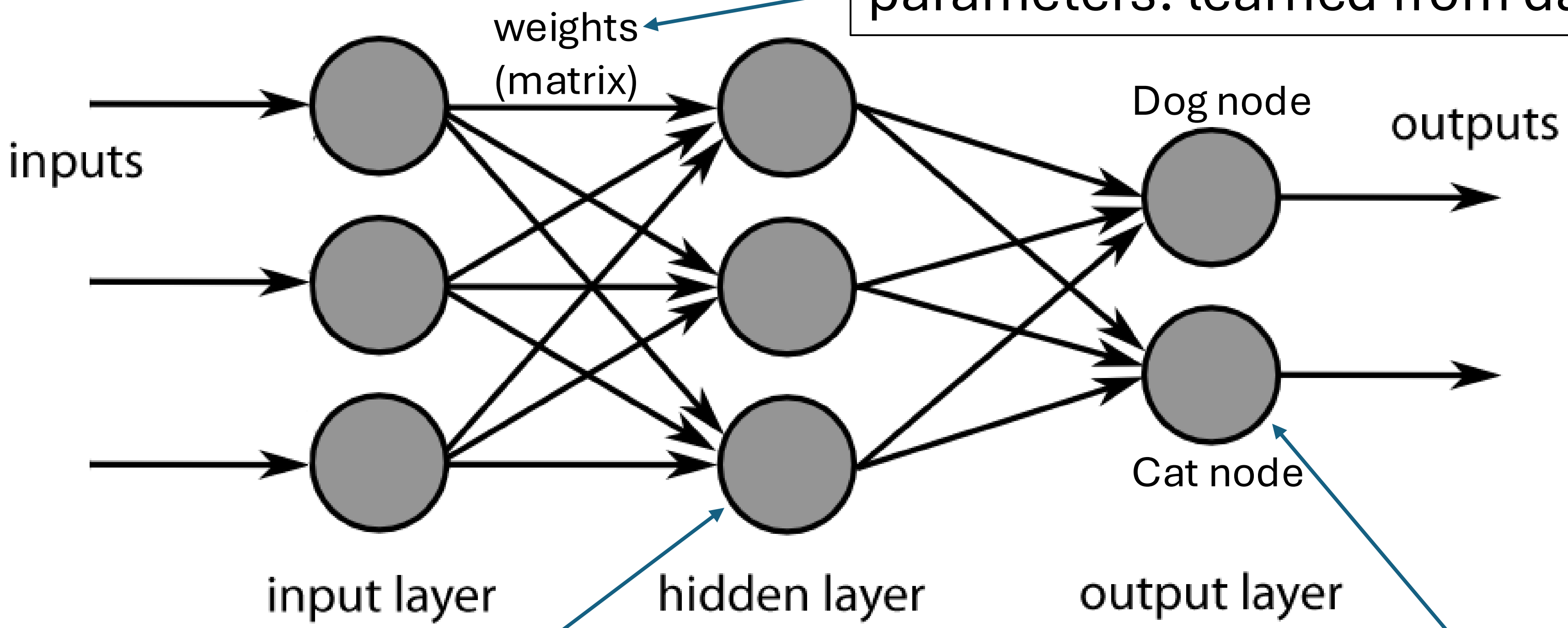


Neural Network as Nonlinear Function



Neural Network as Nonlinear Function

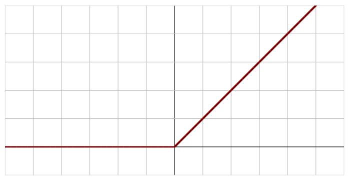
parameters: learned from data



e.g., node 3

probability for „cat“
(softmax instead of activation function)

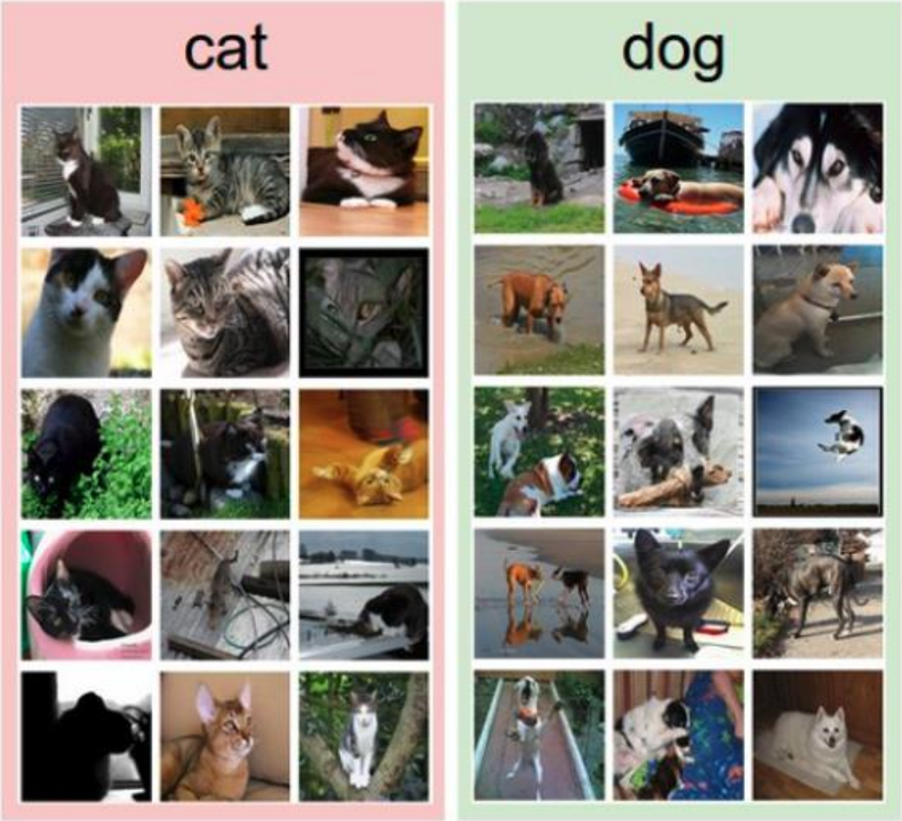
linear transformation
with nonlinear
activation functions



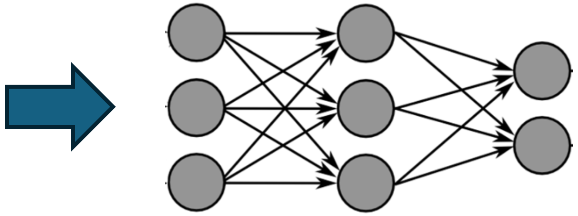
$$\left(\overrightarrow{w_3} \cdot \vec{x}\right)$$

Training with Labeled Images

label:



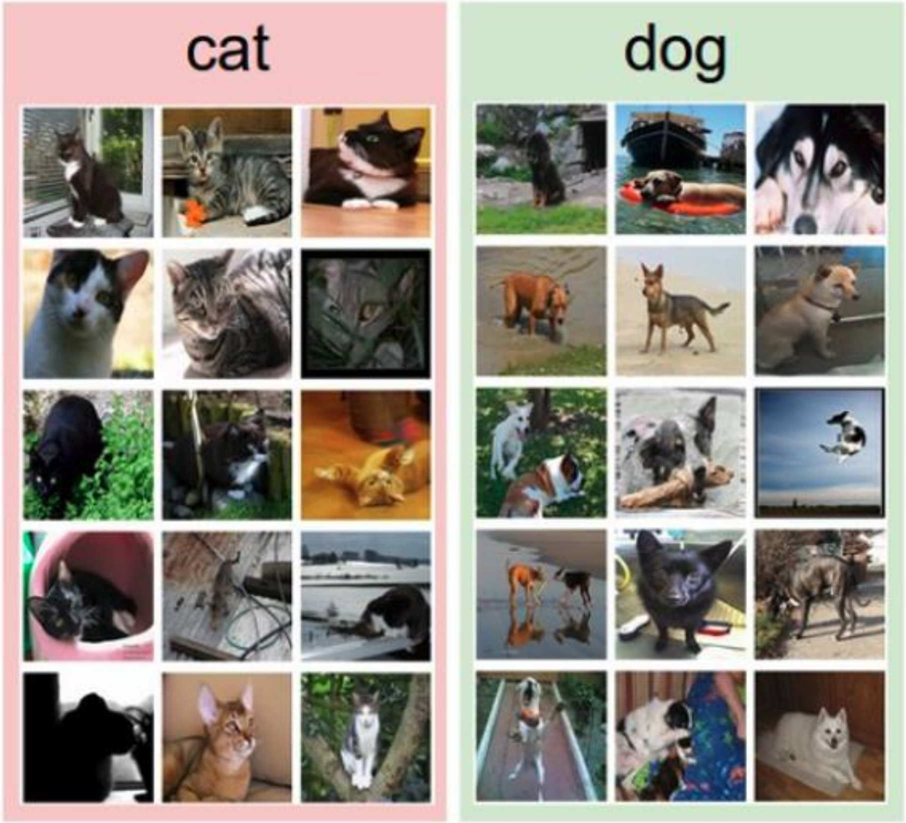
forward pass

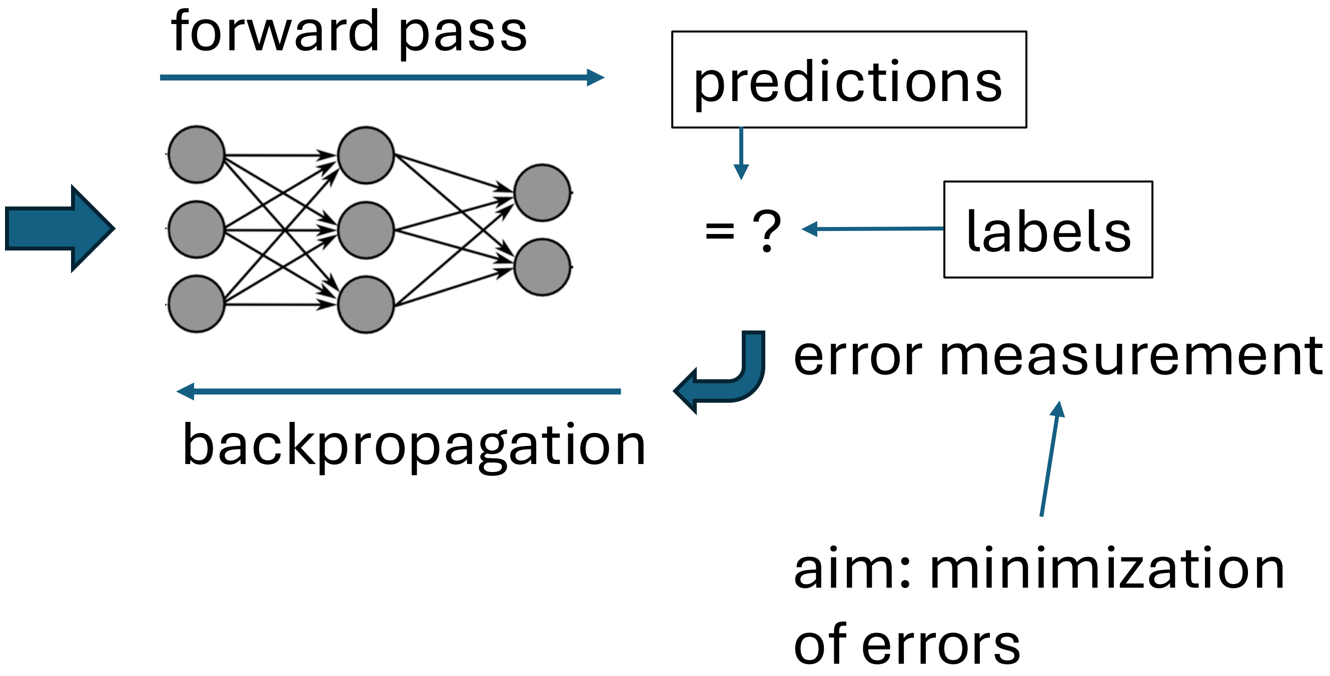


predictions

Training with Labeled Images

label:



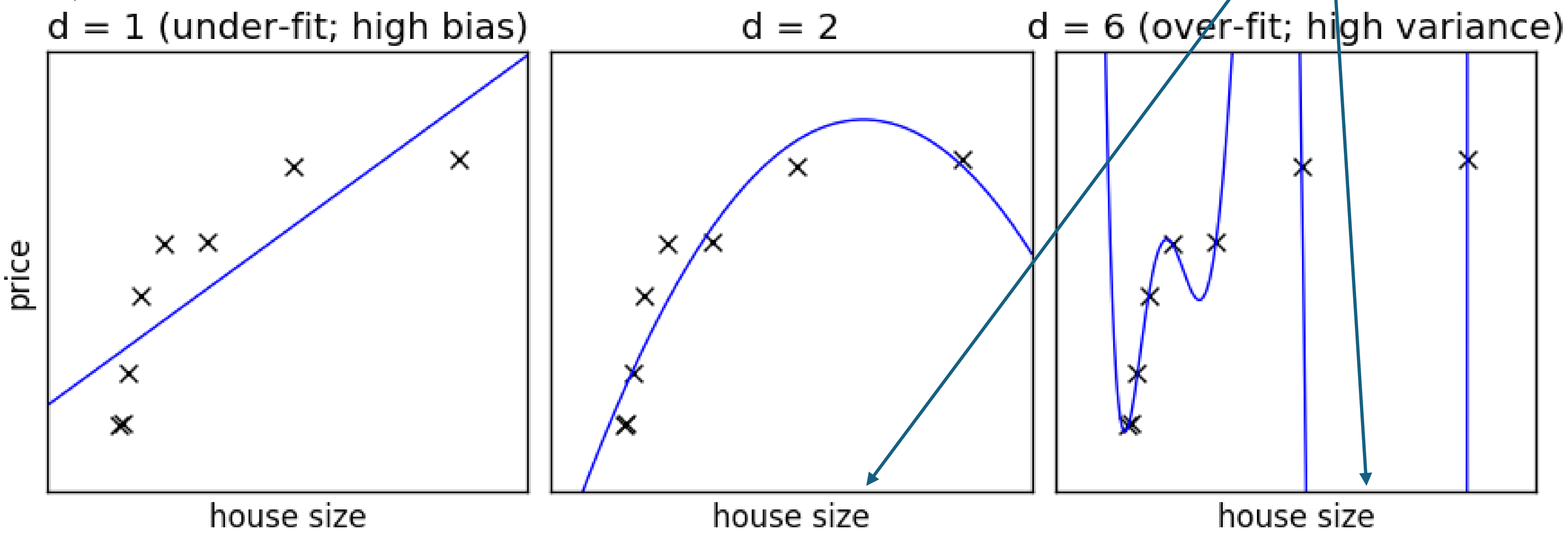


The Training Problem: Under- and Over-Fitting

example: polynomial fit

need to predict new samples (not in training data set): interpolation

degree of fitted polynomial



Training: Iterative Adjustment of Weights

```
for epoch in range(epochs):
    for X, y in mini_batches:
        pred = model(X)
        cost = loss(pred, y)
        cost.backward()
        optimizer.step()
```

quantification of error (here: cross entropy)

chain rule to calculate gradient (direction) of cost function with respect to weights

one step in direction of steepest descent: (stochastic) gradient descent

Classification Networks

cross-entropy loss (one output node for each class k):

$$L(y, \hat{y}) = -\sum_{k=1}^{K} y_k \log \hat{y}_k$$

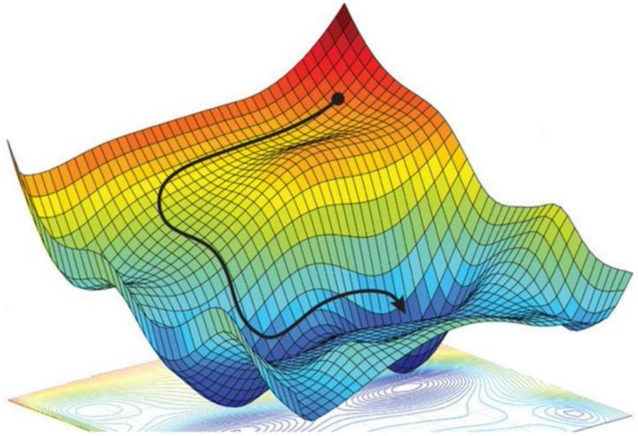
using softmax on output nodes:

$$\hat{y}_k = \frac{e^{o_k}}{\sum_{l=1}^{K} e^{o_l}}$$

regression networks:

- squared error loss: $L(y, \hat{y}) = (y - \hat{y})^2$
- just one output node
- no function on output

Stochastic Gradient Descent (SGD)



step size   cost function

weight updates: $\hat{\boldsymbol{w}} \leftarrow \hat{\boldsymbol{w}} - \eta \nabla_{\hat{\boldsymbol{w}}} J(\hat{\boldsymbol{w}})$

options:

all training examples

- $J(\hat{\boldsymbol{w}}) = \frac{1}{n}\sum_{i=1}^{n} J_i(\hat{\boldsymbol{w}})$ batch gradient descent (whole training dataset)
- $J(\hat{\boldsymbol{w}}) = J_i(\hat{\boldsymbol{w}})$ stochastic gradient descent (single example)
- $J(\hat{\boldsymbol{w}}) = \frac{1}{m}\sum_{i=1}^{m} J_i(\hat{\boldsymbol{w}})$ mini-batch stochastic gradient descent

mini-batch size → hyperparamter: tradeoffs between runtime & memory, convergence & generalization

random fluctuations of SGD help to escape saddle points

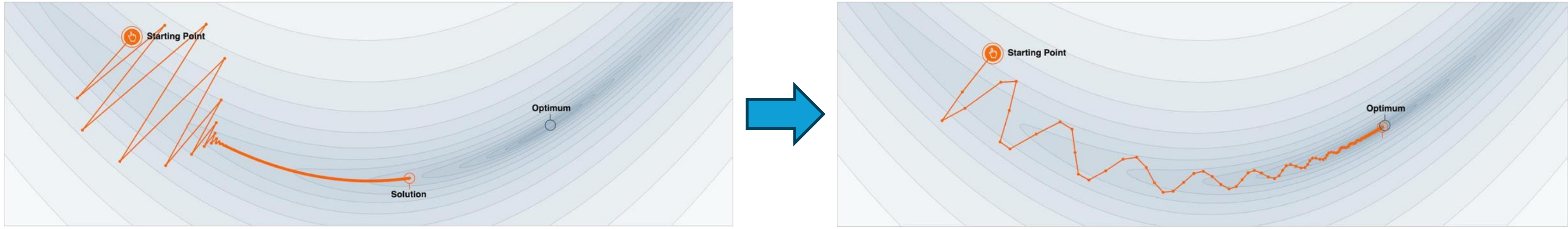
Gradient Descent with Momentum

hyperparameter $(0 < \alpha < 1)$ specifying exponential decay

algorithm:

- estimate gradient: $\boldsymbol{g} = \nabla_{\widehat{\boldsymbol{w}}} J(\widehat{\boldsymbol{w}})$
- compute velocity update: $\boldsymbol{v} \leftarrow \alpha\, \boldsymbol{v} - \eta\, \boldsymbol{g}$
- apply parameter update: $\widehat{\boldsymbol{\theta}} \leftarrow \widehat{\boldsymbol{\theta}} + \boldsymbol{v}$

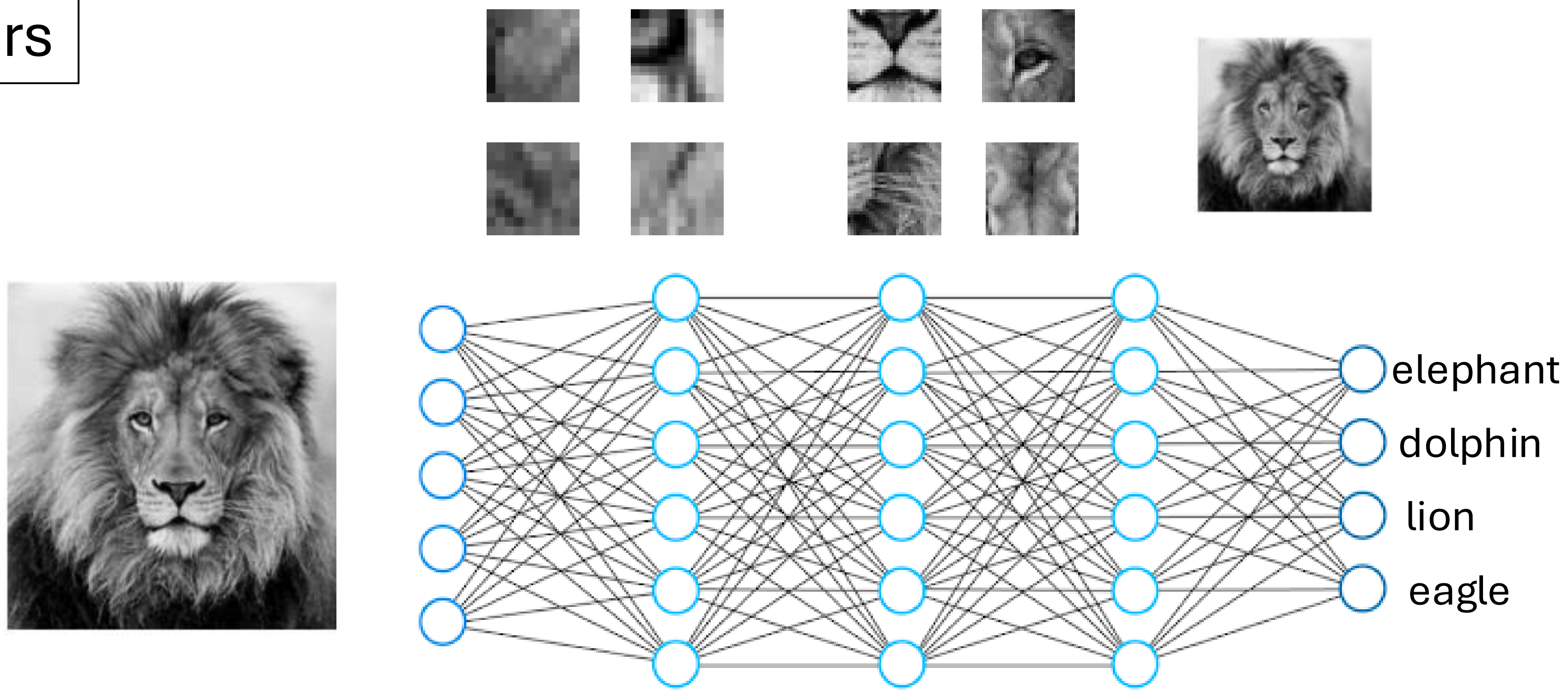
→ no bouncing around of parameters, escape from local minima and saddle points



Hierarchical Representation

increasing abstraction

deep learning:
several hidden layers



Side Note: Inductive Bias for Vision

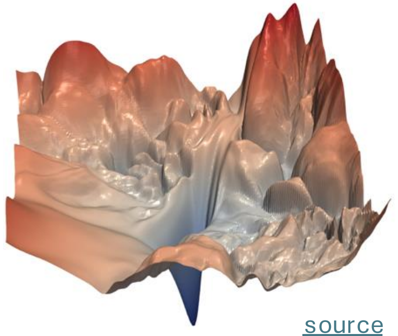
need for consideration of spatial structure

- convolutional neural networks

- vision transformer

How to Train Deep Neural Networks Effectively?

typical loss surface:



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optimization and regularization difficult

- local vs global minima, saddle points, easily overfitting
- many hyperparameters to tune

but many methods to get it working in practice (despite partly patchy theoretical understanding)

- optimization: activation and loss functions, weight initialization, adaptive learning rate (e.g., Adam), learning rate schedulers
- explicit regularization: weight decay, dropout, data augmentation
- implicit regularization: early stopping, batch/layer normalization, SGD

Next: Transformer

recent hype around LLMs

fully started with ChatGPT release end of 2022

technical backbone: transformer architecture

transformers also applicable to other areas, for example computer vision (as alternative to convolutional neural networks)